

PROJECT 2

EXPLORATORY DATA ANALYTICS (EDA) REPORT

The Online Retail Analysis was conducted using Exploratory Data Analysis (EDA) techniques in Excel to uncover trends, identify anomalies, and generate business insights.

STEPS TAKEN FOR EDA

STEP 1: DESCRIPTIVE STATISTICS OVERVIEW

- Numerical Columns were extracted from table and placed in a new sheet.
- Two methods were applied:
 - a. Basic Method: Done manually using basic Excel Functions.
 - b. Advance Method: Done with the aid of Data Analysis (Descriptive Statistics).
 - ❖ Data Tab – Data Analysis – Descriptive Statistics.
 - c. In summary, both methods derived the following:
 - Quantity: Average purchase quantity is 2.95 units, most occurring number is 1, max 5, min 1, median 5.
 - Unit Price: Mean price is approximately \$356, but highly variable. Prices range widely from \$11 to \$700, indicating diverse product categories.
 - Items in Cart: Has an average of 5.5 items, with most customers buying between 1–10 items.
 - Total Price: Mean Revenue = \$1,054. Some transactions are extreme outliers, with totals up to \$3,456.

STEP 2: PRODUCT SALES INSIGHTS AND CONDITIONAL FORMATTING

- Pivot Tables were created from cleaned dataset.
 - ❖ Insert Tab – Pivot Table – From Table/Range
- Insights were summarized using these Pivot Tables
 - Top sellers by revenue: Chairs (\$195,620), Printers (\$195,613), and Laptops (\$192,127).
 - Lowest revenue: Phones (\$151,722).
 - Sales are relatively balanced across categories, but high-ticket items (Laptop, Printer) drive revenue concentration.
- Payment Method Distribution
 - Most used: Credit Card (234 orders, \$263,848 revenue) and Online (258 orders, \$262,443 revenue), although Cash approached Credit Card, getting more orders (246 orders), revenue was underperformed in comparison.
 - Least used: Debit Card (\$232,361).
 - Payment channels are evenly distributed, suggesting no strong customer preference bias.
- Order Status
 - Orders are spread across statuses: Cancelled (250), Returned (247), Pending (237), Shipped (235), Delivered (231).
 - High cancellation/return rates may indicate fulfillment or product satisfaction issues.
- Marketing Source Effectiveness
 - Top channel: Instagram (\$275,285 revenue, 259 orders).
 - Strong performers: Email (\$261,809) and Google (\$250,441).
 - Weakest: Referral (\$226,816).
 - Social media (Instagram, Facebook) drives significant sales, but referrals underperform.

STEP 3: OUTLIER ANALYSIS

- Quartile bounds: $Q1 = \$410.52$, $Q3 = \$1,578.48$, $IQR = \$1,167.96$.
- Upper bound for normal transactions: $\$1,751.93$.
- Transactions above this, for example, $\$3,456$ are flagged as outliers.
- These outliers suggest promotional stacking or bulk buying behavior.

CONCLUSION

- a) Sales are evenly distributed across products and payment methods, but Instagram dominates marketing ROI.
- b) High cancellation and return rates need to be investigated.
- c) Outliers in total price highlight bulk purchases with coupons, which may distort average revenue metrics.
- d) The dataset shows balanced customer behavior with pockets of extreme values worth deeper analysis.

Benedicta Izegbue

benedictaizegbue6@gmail.com